CLOUD NATIVE

```
PS C:\Users\NIKITHA\kitchen-app> kubectl get nodes
NAME                              STATUS   ROLES           AGE   VERSION
kitchen-cluster-control-plane     Ready    control-plane   8d    v1.35.1
PS C:\Users\NIKITHA\kitchen-app> kubectl apply -f k8s/namespace.yaml
namespace/kitchen-app unchanged
PS C:\Users\NIKITHA\kitchen-app>
PS C:\Users\NIKITHA\kitchen-app> kubectl create configmap api-config --from-literal=PORT=3000 -n kitchen-app --dry-run=client -o yaml | kubectl apply -f -
configmap/api-config unchanged
PS C:\Users\NIKITHA\kitchen-app> kubectl create secret generic mongodb-secret `
>>   --from-literal=mongo-root-username=admin `
>>   --from-literal=mongo-root-password=Secret123! `
>>   --from-literal=mongo-uri='mongodb://admin:Secret123!@mongodb-service.kitchen-app.svc.cluster.local:27017/kitchendb?authSource=admin' `
>>   -n kitchen-app --dry-run=client -o yaml | kubectl apply -f -
secret/mongodb-secret unchanged
PS C:\Users\NIKITHA\kitchen-app>
PS C:\Users\NIKITHA\kitchen-app> kubectl apply -f k8s/db/statefulset.yaml
service/mongodb-service unchanged
statefulset.apps/mongodb configured
PS C:\Users\NIKITHA\kitchen-app> kubectl apply -f k8s/api/deployment.yaml
service/api unchanged
service/api-nodeport unchanged
deployment.apps/api-deployment unchanged
PS C:\Users\NIKITHA\kitchen-app> kubectl apply -f k8s/web-ui/deployment.yaml
configmap/web-ui-config unchanged
deployment.apps/web-ui-deployment unchanged
service/web-ui-service unchanged
PS C:\Users\NIKITHA\kitchen-app>
PS C:\Users\NIKITHA\kitchen-app> kubectl apply -f k8s/monitoring/prometheus-config.yaml
configmap/prometheus-config unchanged
deployment.apps/prometheus-deployment unchanged
service/prometheus-service unchanged
PS C:\Users\NIKITHA\kitchen-app> kubectl apply -f k8s/monitoring/grafana-dashboard.yaml
configmap/grafana-dashboards unchanged
PS C:\Users\NIKITHA\kitchen-app> kubectl apply -f k8s/monitoring/grafana-deployment.yaml
configmap/grafana-datasources unchanged
deployment.apps/grafana-deployment unchanged
service/grafana-service unchanged
PS C:\Users\NIKITHA\kitchen-app> kubectl apply -f k8s/monitoring/prometheus-config.yaml
configmap/prometheus-config unchanged
deployment.apps/prometheus-deployment unchanged
service/prometheus-service unchanged
```

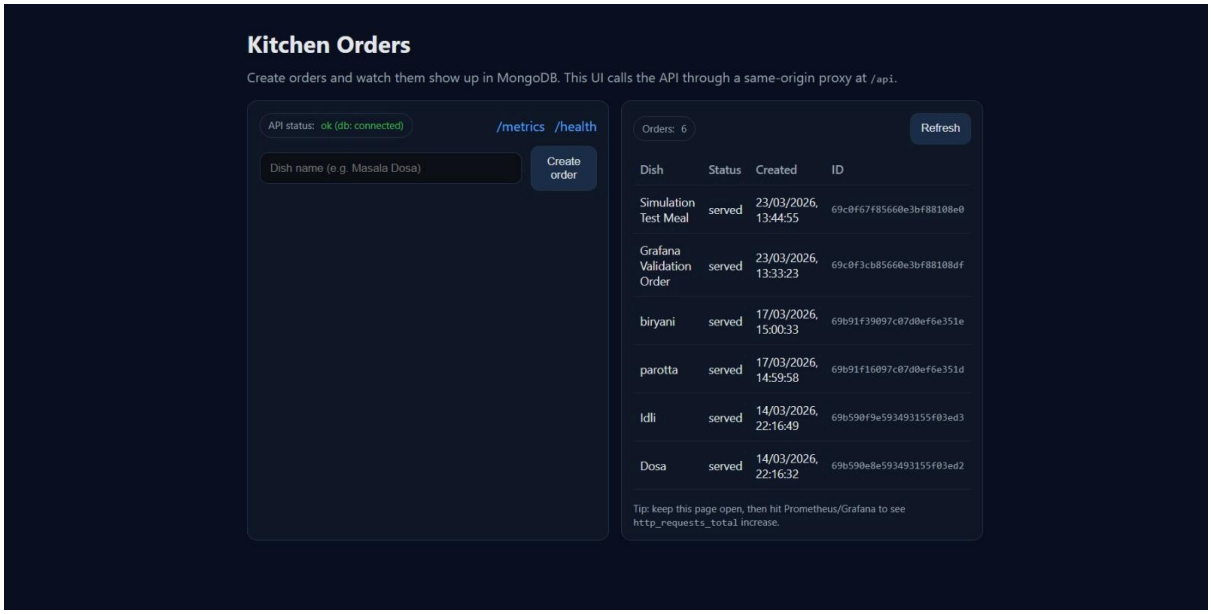
```
PS C:\Users\NIKITHA\kitchen-app> kubectl port-forward -n kitchen-app svc/grafana-service 3000:3000
>>
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
```

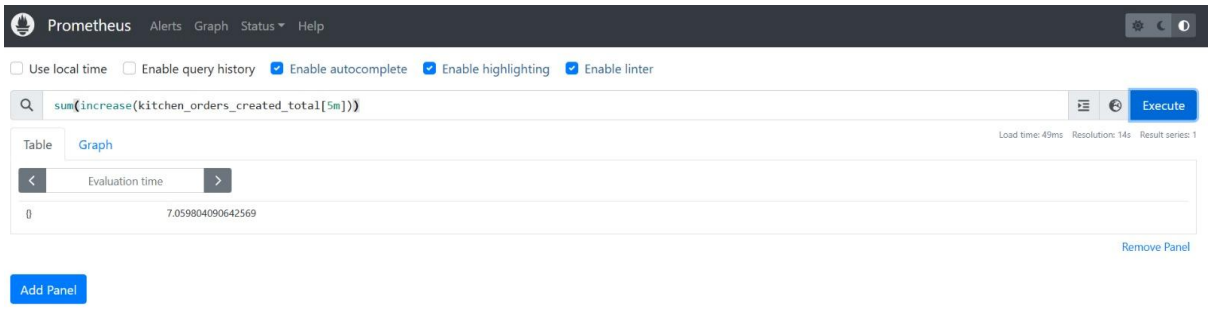
```
PS C:\Users\NIKITHA\kitchen-app> kubectl port-forward -n kitchen-app svc/api 3001:3000
>>
Forwarding from 127.0.0.1:3001 -> 3000
Forwarding from [::1]:3001 -> 3000
Handling connection for 3001
```

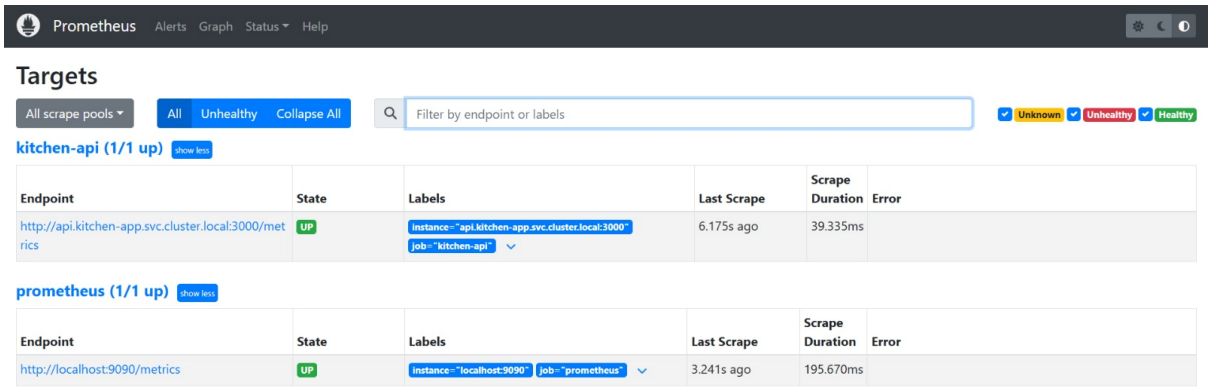
```
PS C:\Users\NIKITHA\kitchen-app> Invoke-RestMethod -Method Post -Uri "http://localhost:30000/orders"

PS C:\Users\NIKITHA\kitchen-app> Invoke-RestMethod -Method Post -Uri "http://localhost:3001/orders" -
ContentType "application/json" -Body (@{ dish = "Grafana Validation Order" } | ConvertTo-Json)
>> Invoke-RestMethod -Method Get -Uri "http://localhost:3001/metrics" | Select-Object -First 5
>>


_id             : 69c0f3cb85660e3bf88108df
dish            : Grafana Validation Order
status          : pending
createdAt       : 2026-03-23T08:03:23.431Z
statusUpdatedAt : 2026-03-23T08:03:23.431Z
```

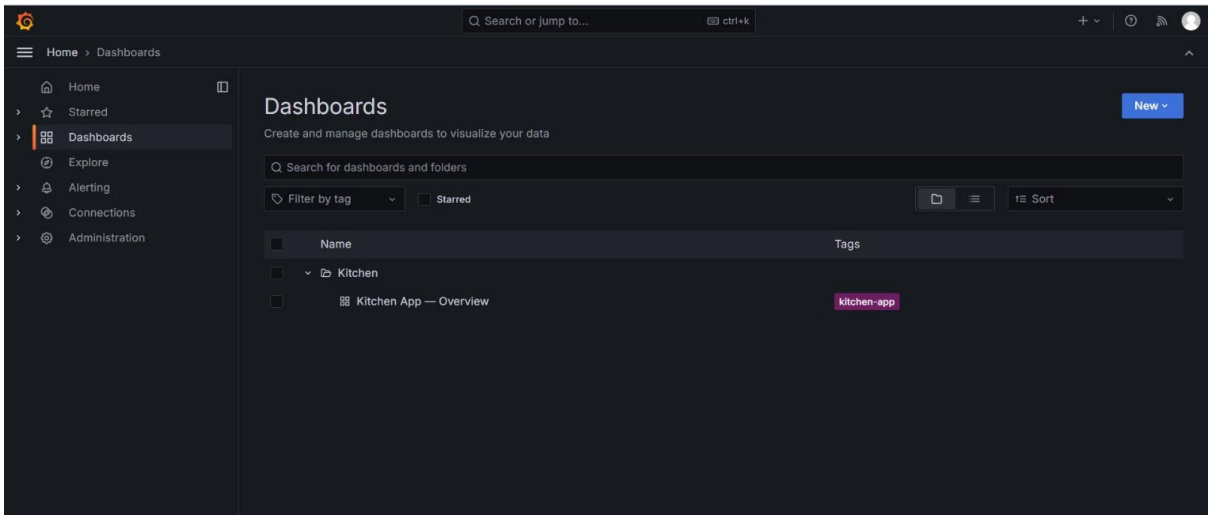
```
PS C:\Users\NIKITHA\kitchen-app> Invoke-RestMethod http://localhost:3001/health

status db
------ --
ok     connected


PS C:\Users\NIKITHA\kitchen-app> Invoke-RestMethod http://localhost:3001/orders


_id             : 69b590e8e593493155f03ed2
dish            : Dosa
status          : served
createdAt       : 2026-03-14T16:46:32.905Z
statusUpdatedAt : 2026-03-14T16:46:47.954Z

_id             : 69b590f9e593493155f03ed3
dish            : Idli
status          : served
createdAt       : 2026-03-14T16:46:49.286Z
statusUpdatedAt : 2026-03-14T16:47:04.925Z

_id             : 69b91f16097c07d0ef6e351d
dish            : parotta
status          : served
createdAt       : 2026-03-17T09:29:58.331Z
statusUpdatedAt : 2026-03-17T09:30:13.585Z
```

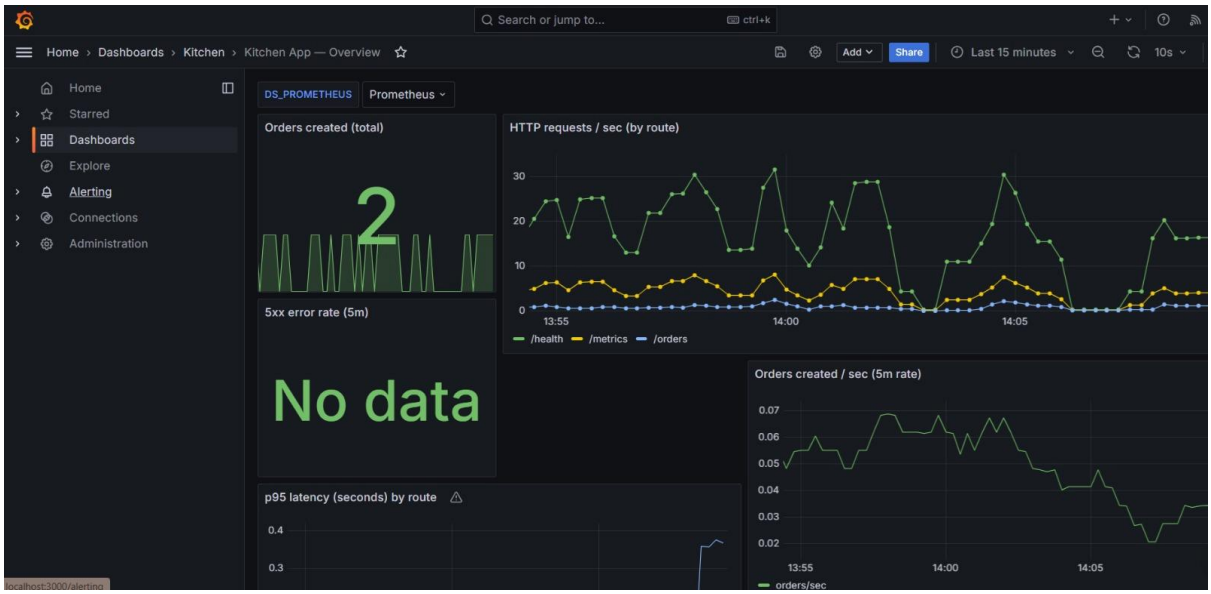
```
PS C:\Users\NIKITHA\kitchen-app> kubectl get pods -n kitchen-app
NAME                                    READY   STATUS    RESTARTS       AGE
api-deployment-6b665b894-7d4jb          1/1     Running   18 (33m ago)   8d
api-deployment-6b665b894-m6zl2          1/1     Running   17 (48m ago)   8d
grafana-deployment-84dc8fd97f-w6nqq     1/1     Running   0              28m
mongodb-0                               1/1     Running   3 (50m ago)    8d
prometheus-deployment-59f9cbbd85-fgcf2  1/1     Running   1 (33m ago)    107m
web-ui-deployment-7b845d5556-8k88p      1/1     Running   12 (49m ago)   8d
web-ui-deployment-7b845d5556-wfz2j      1/1     Running   15 (49m ago)   8d
PS C:\Users\NIKITHA\kitchen-app> kubectl get svc -n kitchen-app
NAME                 TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
api                  ClusterIP   10.96.236.176   <none>        3000/TCP         8d
api-nodeport         NodePort    10.96.147.43    <none>        3000:30000/TCP   8d
grafana-service      ClusterIP   10.96.172.8     <none>        3000/TCP         103m
mongodb-service      ClusterIP   None            <none>        27017/TCP        8d
prometheus-service   ClusterIP   10.96.23.91     <none>        9090/TCP         107m
web-ui-service       NodePort    10.96.175.228   <none>        80:30080/TCP     8d
```


Kitchen Orders
Create orders and watch them show up in MongoDB. This UI calls the API through a same-origin proxy at /api.
API status: ok (db: connected)
/metrics /health
Dish name (e.g. Masala Dosa)
Create order
Orders: 6
Refresh
Dish Status Created ID
Simulation Test Meal served 23/03/2026, 13:44:55 69c0f67f85660e3bf88108e0
Grafana Validation Order served 23/03/2026, 13:33:23 69c0f3cb85660e3bf88108df
biryani served 17/03/2026, 15:00:33 69b91f39097c07d0ef6e351e
parotta served 17/03/2026, 14:59:58 69b91f16097c07d0ef6e351d
Idli served 14/03/2026, 22:16:49 69b590f9e593493155f03ed3
Dosa served 14/03/2026, 22:16:32 69b590e8e593493155f03ed2
Tip: keep this page open, then hit Prometheus/Grafana to see http_requests_total increase.


Prometheus Alerts Graph Status Help
Use local time
Enable query history
Enable autocomplete
Enable highlighting
Enable linter
sum(increase(kitchen_orders_created_total[5m]))
Execute
Table Graph
Load time: 49ms Resolution: 14s Result series: 1
Evaluation time
{} 7.059804090642569
Remove Panel
Add Panel


Prometheus Alerts Graph Status Help
Targets
All scrape pools
All Unhealthy Collapse All
Filter by endpoint or labels
Unknown Unhealthy Healthy
kitchen-api (1/1 up) show less
Endpoint State Labels Last Scrape Scrape Duration Error
http://api.kitchen-app.svc.cluster.local:3000/metrics UP instance="api.kitchen-app.svc.cluster.local:3000" job="kitchen-api" 6.175s ago 39.335ms
prometheus (1/1 up) show less
Endpoint State Labels Last Scrape Scrape Duration Error
http://localhost:9090/metrics UP instance="localhost:9090" job="prometheus" 3.241s ago 195.670ms


Search or jump to...
ctrl+k
Home › Dashboards
Home
Starred
Dashboards
Explore
Alerting
Connections
Administration
Dashboards
New
Create and manage dashboards to visualize your data
Search for dashboards and folders
Filter by tag
Starred
Sort
Name
Tags
Kitchen
Kitchen App — Overview
kitchen-app


Search or jump to...
ctrl+k
Home › Dashboards › Kitchen › Kitchen App — Overview
Add
Share
Last 15 minutes
10s
Home
Starred
Dashboards
Explore
Alerting
Connections
Administration
DS_PROMETHEUS
Prometheus
Orders created (total)
2
HTTP requests / sec (by route)
30
20
10
0
13:55
14:00
14:05
/health
/metrics
/orders
5xx error rate (5m)
No data
Orders created / sec (5m rate)
0.07
0.06
0.05
0.04
0.03
0.02
13:55
14:00
14:05
orders/sec
p95 latency (seconds) by route
0.4
0.3

Search or jump to...
ctrl+k
Home › Dashboards › Kitchen › Kitchen App — Overview
Add
Share
Last 15 minutes
10s
Home
Starred
Dashboards
Explore
Alerting
Connections
Administration
p95 latency (seconds) by route
0.4
0.3
0.2
0.1
0
14:10
14:15
14:20
/health
/metrics
/orders
API RSS memory (bytes)
100000000
90000000
80000000
70000000
60000000
50000000
14:10
14:15
14:20
rss
rss
0.05
0.04
0.03
0.02
14:10
14:15
14:20
orders/sec
API CPU (seconds/sec)
7
6
5
4
3
2
14:10
14:15
14:20
cpu